Vision-Language Models for Embodied Reasoning: A Survey

Vision-language models are becoming a central interface for embodied reasoning because they connect visual perception, language instructions, spatial concepts, object affordances, and action selection. This survey reviews vision-language and vision-language-action models for embodied agents, robotics, navigation, manipulation, instruction following, environment feedback, world modeling, training, benchmarks, and safety. We synthesize an arXiv-derived corpus of 300 papers collected during this run and organize the literature around embodied VLM architectures, robot manipulation, navigation and embodied QA, task planning, multimodal grounding, simulation and feedback, imitation and reinforcement learning, and evaluation. The survey argues that embodied reasoning differs from static image-language understanding because predictions must be grounded in physical state, action consequences, temporal memory, and safety constraints. We close with prospects for robust visual grounding, open-vocabulary affordances, long-horizon planning, simulation-to-real transfer, and human-aligned embodied agents.

Keywords:

vision-language models; embodied reasoning; robotics; vision-language-action models; navigation; manipulation; multimodal grounding

1 Introduction

Embodied reasoning asks an AI system to connect perception, language, memory, and action in a physical or simulated environment. Vision-language models (VLMs) are attractive for this setting because they can interpret images, follow instructions, name objects, explain scenes, and use broad visual-language priors. Yet embodiment changes the problem: a model's output may become a robot action, a navigation decision, or a long-horizon plan whose consequences unfold over time [136, 142, 150, 171, 194, 211, 265, 299]. Static recognition is not enough; the system must reason about affordances, spatial relations, uncertainty, safety, and feedback.

Recent embodied systems increasingly combine VLMs with action policies, planners, simulators, object detectors, retrievers, memory modules, and robot controllers. Vision-language-action (VLA) models attempt to map visual observations and language goals directly into actions, while modular agents use VLMs for perception, grounding, subgoal selection, or failure diagnosis [17, 136, 142, 150, 183, 194, 201, 299]. Both styles face the same core challenge: the semantic world described in language must be aligned with the physical world in which actions have costs and risks.

This survey reviews VLMs for embodied reasoning from a systems perspective. We focus on how models represent embodied state, infer affordances, plan over time, use environment feedback, learn from demonstrations, and evaluate task success. The central claim is that embodied VLMs require a closed loop: observe, ground, plan, act, evaluate, and revise.

2 Background

2.1 Embodied Reasoning

Embodied reasoning differs from ordinary visual question answering because the agent is situated. It observes from a particular viewpoint, acts with limited effectors, changes the environment, and must respect physical constraints. Language instructions are often underspecified: "put the mug near the sink" requires object recognition, spatial relation reasoning, grasp planning, and safe motion. VLMs provide semantic priors, but they need action grounding and state tracking to become embodied controllers [136, 142, 150, 171, 183, 194, 201, 211].

2.2 Vision-Language-Action Models

VLA models extend VLMs toward action. They condition on images, language, and sometimes proprioception or history, then produce robot actions, waypoints, skills, or subgoals. Some systems train end-to-end policies; others use VLMs to select from a library of skills. The key tension is abstraction: high-level language reasoning helps generalization, while low-level control requires precise timing, geometry, and safety [17, 136, 142, 150, 183, 194, 201, 299].

2.3 Embodied Benchmarks and Environments

Embodied research relies on simulated homes, navigation environments, manipulation suites, real robot platforms, and instruction-following datasets. Simulation enables scale and reproducibility, but real-world deployment exposes sensor noise, dynamics, clutter, and safety constraints. A benchmark should therefore clarify whether it measures perception, language grounding, planning, control, or recovery from failure [17, 59, 142, 171, 183, 201, 250, 293].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with vision-language models, embodied reasoning, embodied agents, large multimodal models, robotics, vision-language- action models, robot manipulation, robot planning, navigation, embodied question answering, spatial reasoning, affordances, instruction following, world models, simulation, environment feedback, household robotics, open-vocabulary robotics, SayCan, RT-2, PaLM-E, OpenVLA, ALFRED, and Habitat terms. Records were retained when title or abstract evidence connected language, vision-language, or multimodal models with embodied agents, robotics, navigation, manipulation, spatial reasoning, grounding, training, evaluation, or safety. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize embodied VLM methods along six axes. The first is *embodiment*: mobile agent, manipulator, mobile manipulator, simulated household agent, autonomous vehicle, or general interactive environment. The second is *state representation*: image, video, depth, object graph, map, language memory, proprioception, or action history. The third is *reasoning role*: perception, grounding, planning, affordance inference, skill selection, failure diagnosis, or policy generation. The fourth is *action interface*: natural-language plan, symbolic action, API call, waypoint, trajectory, or low-level control. The fifth is *learning signal*: web pretraining, instruction tuning, imitation learning, reinforcement learning, self-supervision, or human feedback. The sixth is *evaluation*: task success, trajectory efficiency, generalization, physical safety, robustness, and human trust.

3.3 Embodied VLM and VLA Architectures

Architectures range from modular pipelines to end-to-end policies. Modular systems use a VLM as a semantic interpreter or planner, then call detectors, mapping systems, motion planners, or skill libraries. VLA systems learn a more direct mapping from multimodal observations to actions. Modular designs are interpretable and easier to constrain; end-to-end models can exploit large-scale data and reduce interface mismatch. Many practical systems combine both approaches [17, 136, 142, 150, 183, 194, 201, 299].

3.4 Manipulation and Affordance Grounding

Manipulation requires reasoning about what can be done with objects. A cup can be grasped by its handle, a drawer can be pulled, and a fragile object should be moved carefully. VLMs help by associating object categories with affordances and task goals, but robot control requires geometry, contact, and uncertainty estimates. Systems therefore pair VLMs with grasp detectors, object segmentation, skill primitives, and feedback from failed actions [17, 136, 142, 171, 194, 201, 211, 265].

3.5 Navigation, Embodied QA, and Spatial Reasoning

Navigation tasks require an agent to connect language instructions with spatial layout. Vision-language navigation, embodied question answering, and exploration benchmarks test whether a model can identify goals, maintain a map, reason about unseen areas, and decide where to move next. VLMs provide open-vocabulary scene understanding, but spatial memory and action history are essential for avoiding loops and recovering from wrong turns [171, 193, 201, 211, 250, 265, 279, 299].

3.6 Instruction Following and Task Planning

Embodied instructions often specify goals rather than low-level actions. A household task may involve finding objects, opening containers, using tools, and sequencing subtasks over many minutes. VLMs can decompose instructions and monitor progress, but long-horizon planning needs explicit state, preconditions, and failure handling. Language planners are most useful when their plans are checked by environment feedback or executable skills [136, 142, 150, 171, 194, 211, 265, 299].

3.7 Multimodal Grounding and Perception

Embodied reasoning depends on grounding words in perceived entities and actions. Open-vocabulary detectors, segmenters, scene graphs, and visual grounding models allow agents to connect language to objects and regions. The difficulty is that grounding must remain stable across viewpoint changes, occlusion, object movement, and interaction. Robust embodied VLMs therefore need persistent object identity and uncertainty-aware perception [136, 142, 150, 171, 183, 194, 201, 211].

3.8 World Models, Simulation, and Feedback

World models predict how the environment changes after actions. They can support lookahead, counterfactual reasoning, and safer planning. Simulators provide cheap feedback for training and evaluation, but embodied agents must eventually handle real-world discrepancies. Feedback from execution, perception, and human correction can help VLMs revise plans and calibrate confidence [136, 142, 150, 171, 183, 194, 211, 265].

3.9 Training and Adaptation

Embodied VLMs draw on web-scale visual-language data, robot demonstrations, simulated trajectories, video, and language instructions. Imitation learning teaches skills from demonstrations; reinforcement learning optimizes task success; instruction tuning improves language following; and offline datasets provide broad coverage. The open problem is combining semantic generalization with reliable physical control [17, 150, 183, 193, 201, 250, 293, 299].

3.10 Benchmarks, Evaluation, and Safety

Evaluation must measure more than final answer accuracy. Embodied systems need task success, collision rate, path efficiency, manipulation success, recovery behavior, generalization to new objects and scenes, and safe handling of uncertainty. Human-centered evaluation also matters because users must understand what the agent is trying to do and when it is unsure [17, 59, 142, 171, 183, 201, 250, 293].

3.11 Representative Literature Map

The following map cites the full retained corpus by theme, making the survey coverage explicit.

Vision-language-action models and embodied agents The vision-language-action models and embodied agents cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [71, 80, 124, 143, 156, 175, 236, 250, 279]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [108, 129, 158, 169, 177, 181, 232, 269, 287]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [20, 31, 52, 82, 87, 174, 271, 289, 297]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [27, 149, 202, 206, 296].

Robot manipulation and affordance grounding The robot manipulation and affordance grounding cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [141, 142, 201, 211, 256, 265, 267, 293, 299]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [56, 65, 78, 114, 119, 135, 192, 218, 294]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [11, 23, 44, 53, 95, 127, 153, 179, 230]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [16, 58, 61, 68, 116, 166, 212, 227, 237]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [1, 32, 46, 126, 139, 187, 195, 204, 253]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [50, 57, 157, 205, 217, 231, 262, 284, 286]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [70, 74, 132, 161, 209, 226, 238, 243, 298]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [47, 98, 130, 203, 210, 224, 241, 244, 300]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [15, 37, 86, 99, 140, 152, 199, 233, 239]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [45, 60, 63, 93, 110, 155, 160, 220, 283]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [2, 35, 54, 100, 120, 191, 258, 273, 282]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [18, 104, 196].

Navigation, embodied qa, and spatial reasoning The navigation, embodied qa, and spatial reasoning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [39, 85, 113, 125, 167, 168, 173, 261, 277]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [19, 77, 109, 146, 198, 222, 223, 288, 295]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [151].

Instruction following and task planning The instruction following and task planning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [9, 17, 40, 62, 150, 176, 182, 255, 278]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [14, 29, 88, 102, 112, 118, 186, 229, 249]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [94, 96, 103, 123, 245, 257, 264, 275].

Multimodal perception and grounding The multimodal perception and grounding cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [59, 92, 136, 171, 185, 194, 208, 225, 247]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [12, 38, 105, 138, 145, 148, 170, 200, 266]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [3, 43, 51, 67, 84, 154, 235, 263, 276]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [5, 10, 26, 34, 36, 64, 81, 162, 216]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [6, 7, 25, 28, 49, 76, 106, 144, 213]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [33, 41, 66, 89, 131, 188–190, 281]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [72, 107, 137, 147, 159, 219, 242, 260, 280]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [73, 163].

World models, simulation, and environment feedback The world models, simulation, and environment feedback cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [8, 75, 97, 172, 214, 274, 292].

Training, imitation, and reinforcement learning The training, imitation, and reinforcement learning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [24, 42, 90, 101, 111, 193, 234, 254, 270]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [48, 79, 91, 134, 165, 272].

Benchmarks, evaluation, and safety The benchmarks, evaluation, and safety cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [22, 115, 121, 164, 183, 184, 207, 246, 268]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [4, 13, 55, 128, 197, 240, 252, 285, 291]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [21, 69, 117, 133, 221, 248, 251, 259, 290]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [30, 83, 122, 178, 180, 215, 228].

4 Challenges and Prospects

4.1 Semantic Grounding Meets Physical Constraint

The strongest VLMs still make mistakes when language concepts must be grounded in precise geometry or force. Future systems need representations that keep semantic labels, object states, affordances, and physical constraints aligned over time.

4.2 Long-Horizon Memory and Recovery

Embodied tasks unfold across many observations and actions. Agents need memory for goals, explored spaces, object locations, failed attempts, and user corrections. Recovery from failure should be a first-class capability rather than a rare benchmark edge case.

4.3 Simulation-to-Real Transfer

Simulation enables scale but can hide physical details. The field needs better sim-to-real protocols, domain randomization, real-world evaluation, and benchmarks that report the gap between simulated and physical success.

4.4 Safety, Oversight, and Human Alignment

Embodied agents can damage objects, enter unsafe spaces, or act against user intent. Safety requires constrained action interfaces, uncertainty estimation, permission checks, human approval for high-risk actions, and transparent explanations of planned behavior.

5 Conclusion

Vision-language models provide powerful semantic priors for embodied reasoning, but embodiment requires more than visual description. This survey reviewed VLM and VLA architectures, manipulation, navigation, instruction following, grounding, world models, training, benchmarks, and safety. The central lesson is that embodied intelligence is closed-loop: models must ground language in perceived state, act through constrained interfaces, learn from feedback, and remain safe under physical uncertainty.

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